Machine Learning in Economics/Finance

Predicting U.S. CPI Inflation Using Machine Learning: ARIMAX, XGBoost, and LSTM

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Abstract

This paper applies a four-stage machine learning pipeline to forecast U.S. Consumer Price Index (CPI) inflation using 450 monthly macroeconomic observations spanning 1987 to 2024. First, LASSO regression is used to select the most informative predictors from a candidate set of ten macroeconomic variables. Second, an ARIMAX(3,1,0) model serves as the traditional time series baseline. Third, XGBoost — evaluated in both baseline and hyperparameter-tuned configurations — acts as the tree-based machine learning challenger. Fourth, a two-layer LSTM neural network serves as the deep learning challenger. Evaluated on an out-of-sample test set covering July 2017 to December 2024 — which includes the extraordinary 2021–2023 inflation surge — XGBoost substantially outperforms all other models, achieving a Mean Absolute Error (MAE) of 0.641 compared to 1.752 for ARIMAX and 0.846 for LSTM. While LSTM captures directional movements well, XGBoost's superior performance stems from its ability to exploit non-linear interactions among lagged inflation and macroeconomic predictors selected by LASSO. Lagged inflation emerges as the dominant predictor across all models, consistent with adaptive expectations theory.

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1 Problem Identification

1.1 Research Question

Can machine learning and deep learning methods outperform traditional time series models in forecasting U.S. consumer price inflation, and which macroeconomic indicators are most predictive of inflation dynamics?

1.2 Background and Motivation

Inflation forecasting is one of the most consequential tasks in macroeconomics. Accurate predictions of the Consumer Price Index (CPI) inform monetary policy decisions by central banks, guide investment strategies for asset managers, and shape fiscal planning for governments. The U.S. Federal Reserve explicitly targets a 2% annual inflation rate, making deviations from this target a signal for policy intervention.

The period from 2020 to 2024 presented an extraordinary forecasting challenge. Following the COVID-19 pandemic, U.S. inflation surged from below 2% to a peak of approximately 9.1% in June 2022 — the highest level since 1981 — before gradually declining. This episode exposed the limitations of traditional linear models, which struggled to anticipate the speed and magnitude of the shock. It raises a key empirical question: can machine learning and deep learning models, which capture non-linear relationships and long-range temporal dependencies, perform better under such conditions?

1.3 Data and Relevance

All macroeconomic data are sourced from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis, widely regarded as the gold standard for U.S. macroeconomic data. Crude oil prices are sourced from FRED's WTI series (MCOILWTICO), which extends back to 1986 and yields 450 monthly observations after feature engineering, spanning July 1987 to December 2024. This long historical coverage is particularly important for training the LSTM model, which requires sufficient sequential data to learn patterns across multiple business cycles. Predictor variables are summarised in Table 1.

Table 1: Data Sources and Variable Descriptions

Variable	**FRED Code**	**Description**	**Role**
CPI	CPIAUCSL	Consumer Price Index	Target
Federal Funds Rate	FEDFUNDS	Policy interest rate	Predictor
Unemployment Rate	UNRATE	Labour market slack	Predictor
M2 Money Supply	M2SL	Broad money growth	Predictor
PCE Price Index	PCEPI	Consumption deflator	Predictor
Consumer Sentiment	UMCSENT	U. Michigan index	Predictor
Industrial Production	INDPRO	Output indicator	Predictor
WTI Crude Oil Price	MCOILWTICO	Crude oil price	Predictor

1.4 Exploratory Data Analysis

Figure 1 presents four exploratory visualisations of the dataset. The top-left panel shows the CPI inflation history from 1987 to 2024, highlighting the 2021–2023 surge and the Fed's 2% target. The dataset captures three major inflationary episodes: the late 1980s, the 2008 commodity price surge, and the 2021–2023 post-pandemic surge, providing rich variation for model training.

The top-right correlation matrix reveals several economically meaningful patterns. Lagged inflation (`inflation_lag1` $= 0.96$, `inflation_lag3` $= 0.84$) is strongly correlated with current inflation, confirming strong serial persistence consistent with adaptive expectations theory. Unemployment is negatively correlated with inflation ($r = -0.44$), consistent with the Phillips Curve. Consumer sentiment is also negatively correlated ($r = -0.43$), reflecting that households feel worse off when inflation is high.

The bottom-left panel shows the right-skewed distribution of inflation, with a mean of approximately 3% and a long right tail driven by the 2022 surge. The bottom-right panel overlays inflation with the federal funds rate, visually confirming the delayed monetary policy response: the Fed began raising rates aggressively in 2022 only after inflation had already peaked, illustrating the reverse causality challenge discussed in Section 3.3.

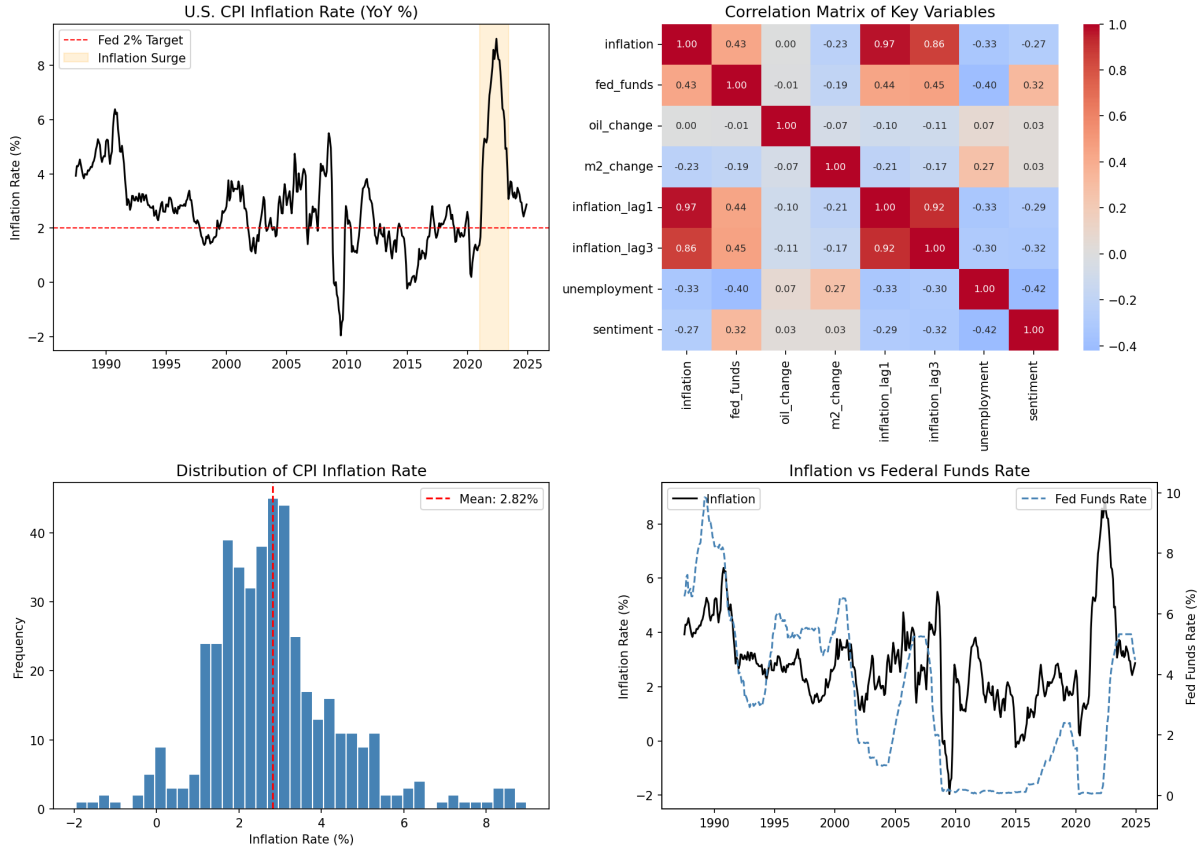

Figure 1: Exploratory Data Analysis (1987–2024). Clockwise from top-left: (1) CPI inflation over time with Fed 2% target and inflation surge highlighted; (2) correlation matrix of key variables; (3) inflation vs. federal funds rate over time; (4) distribution of CPI inflation rate.

2 Application of Machine Learning Tools

2.1 Methodology Overview

This project employs a four-stage pipeline covering multiple course topics. LASSO regression is used for feature selection (Week 2). An ARIMAX model serves as the traditional time series baseline (Week 3). XGBoost acts as the tree-based machine learning challenger (Week 6). Finally, an LSTM neural network serves as the deep learning challenger (Week 9). The data are split chronologically, with 80% used for training (July 1987 – June 2017) and 20% for testing (July 2017 – December 2024).

This is a regression task rather than a classification task, as the target variable — the year-over-year CPI inflation rate — is continuous. Predicting a continuous outcome requires regression methods, whereas classification would only be appropriate if the goal were to predict discrete categories such as whether inflation is above or below a threshold.

2.2 Stage 1 — LASSO for Feature Selection

LASSO (Least Absolute Shrinkage and Selection Operator) regularisation penalises the absolute size of regression coefficients, shrinking irrelevant predictors to exactly zero. This makes it well-suited for feature selection in high-dimensional settings where many candidate variables may be correlated or redundant. LASSO solves the following optimisation problem:

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\} \tag{1}$$

where y_i is the inflation rate, x_{ij} are the standardised predictor variables, β_j are the regression coefficients, and $\lambda \geq 0$ is the regularisation parameter that controls the degree of shrinkage. The optimal λ was selected via 5-fold cross-validation.

Ten candidate features were constructed: the unemployment rate, federal funds rate, M2 monthly growth rate, PCE price index, consumer sentiment index, industrial production growth rate, oil price monthly change, and lagged inflation at 1, 3, and 6 months. All features were standardised using z-score normalisation prior to fitting. Table 2 and Figure 2 present the results.

Table 2: LASSO Regression Coefficients

Feature	**Coefficient**	**Status**
Inflation Lag 1	1.411	Retained
Inflation Lag 3	-0.168	Retained
Oil Price Change	0.132	Retained
Federal Funds Rate	0.115	Retained
Consumer Sentiment	-0.072	Retained
Inflation Lag 6	-0.046	Retained
Unemployment Rate	-0.024	Retained
PCE Price Index	-0.016	Retained
Industrial Production	0.005	Retained
M2 Growth	0.000	Eliminated

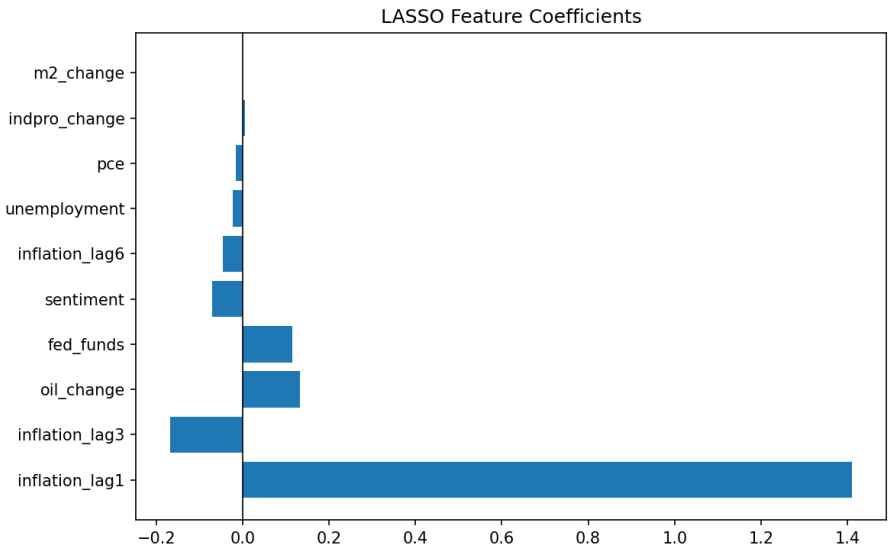

Figure 2: LASSO Feature Coefficients. Only M2 growth is shrunk to zero and excluded from subsequent models.

The dominant role of `inflation_lag1` (coef = 1.411) confirms strong inflationary inertia, consistent with adaptive expectations theory. Only M2 monthly growth was eliminated, while all other nine features were retained. This result suggests that with the expanded 38-year dataset, most macroeconomic variables carry meaningful predictive signal for inflation that LASSO preserves. The retained features are used to inform the construction of the ARIMAX and XGBoost models.

2.3 Stage 2 — ARIMAX Model

ARIMAX (AutoRegressive Integrated Moving Average with eXogenous variables) extends the classical ARIMA framework by incorporating external predictors. It is the standard benchmark for macroeconomic time series forecasting, offering strong interpretability and well-established statistical diagnostics. Table 3 summarises the model specification.

Table 3: ARIMAX Model Specification

Parameter	**Value**
AR order (p)	3
Differencing (d)	1
MA order (q)	0
Exogenous variables	Federal Funds Rate, Oil Price Change, M2 Growth
Selection criterion	AIC
Training sample	July 1987 – June 2017 (360 observations)

The differencing order $d = 1$ addresses the non-stationarity of the inflation series. The model is specified as:

$$\Delta\pi_t = c + \sum_{i=1}^{3} \phi_i \Delta\pi_{t-i} + \beta_1 \text{FedFunds}_t + \beta_2 \text{OilChange}_t + \beta_3 \text{M2Change}_t + \varepsilon_t \quad (2)$$

where π_t denotes the year-over-year CPI inflation rate and Δ denotes first differences. Table 4 presents the estimated coefficients.

Table 4: ARIMAX(3,1,0) Estimation Results

Variable	**Coef.**	**Std. Err.**	**z**	**p-value**
Intercept	-0.001	0.020	-0.074	0.941
Federal Funds Rate	0.719	0.317	2.267	0.023**
Oil Price Change	0.010	0.013	0.772	0.440
M2 Growth	0.003	0.014	0.176	0.860
AR(1)	0.438	0.049	8.885	0.000***
AR(2)	-0.214	0.056	-3.804	0.000***
AR(3)	0.016	0.052	0.309	0.757
σ^2	0.122	0.006	21.963	0.000***
AIC: 279.777 BIC: 310.843 Log-Likelihood: -131.888				
*** p<0.01, ** p<0.05				

2.4 Stage 3 — XGBoost Model

XGBoost (Extreme Gradient Boosting) is an ensemble tree-based method that builds models sequentially, with each tree correcting the residual errors of the previous one. Unlike ARIMAX, XGBoost makes no parametric assumptions about functional form, allowing it to capture non-linearities and interaction effects. Two configurations were evaluated, as summarised in Table 5.

Table 5: XGBoost Model Specifications

Parameter	**Baseline**	**Tuned**
n_estimators	200	200
max_depth	3	3
learning_rate	0.05	0.05
Tuning method	—	GridSearchCV (5-fold CV)
Features	fed_funds, oil_change, m2_change, inflation_lag1, inflation_lag3	

Although M2 growth was eliminated by LASSO under a linear specification, it was retained in the XGBoost feature set as the non-linear model may capture interactions involving M2 that the linear LASSO penalty suppresses. Its inclusion had negligible impact on overall model performance.

The grid search covered `n_estimators` $\in \{100, 200, 300\}$, `max_depth` $\in \{3, 4, 5\}$, and `learning_rate` $\in \{0.01, 0.05, 0.1\}$ with MAE as the scoring metric, achieving a cross-validation MAE of 0.382. Notably, the tuned parameters are identical to the baseline, confirming that the initial hyperparameter choices were already near-optimal. Figure 3 presents the XGBoost feature importance based on information gain.

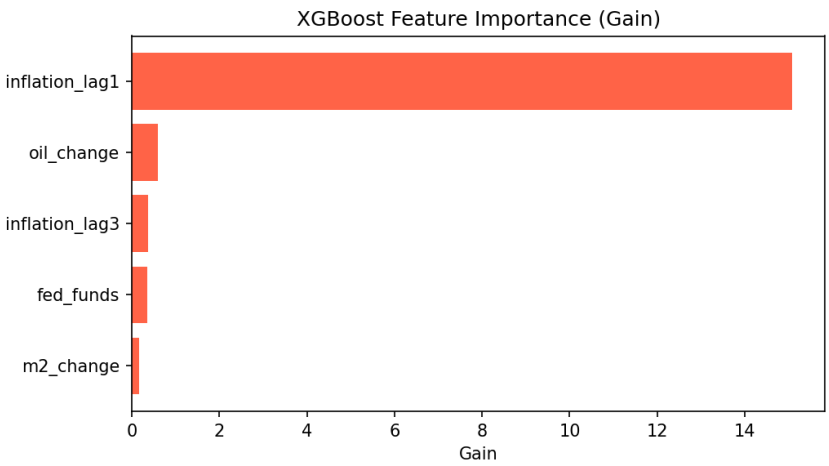

Figure 3: XGBoost Feature Importance (Gain). `inflation_lag1` dominates all other features, providing cross-model validation of the LASSO results.

2.5 Stage 4 — LSTM Model

Long Short-Term Memory (LSTM) networks are a class of recurrent neural networks specifically designed to capture long-range temporal dependencies in sequential data. Unlike ARIMAX and XGBoost, which use fixed lag features, LSTM processes entire sequences of observations and learns to selectively retain or forget information across time steps through its gating mechanism. Table 6 summarises the model specification.

Table 6: LSTM Model Specification

Parameter	**Value**
Framework	PyTorch
Random seed	42 (for reproducibility)
Lookback window	12 months
Layer 1	LSTM, 64 hidden units
Layer 2	LSTM, 32 hidden units
Dropout rate	0.2
Output layer	Fully connected, 1 unit
Optimizer	Adam (lr = 0.001)
Loss function	Mean Squared Error
Early stopping	Patience = 10 epochs (stopped at epoch 77)
Input features	inflation, fed_funds, oil_change, m2_change, unemployment
Training size	348 sequences
Test size	90 sequences

The model uses a 12-month lookback window, meaning the network observes the past 12 months of macroeconomic data to predict the next month's inflation. All input features were standardised prior to training. Dropout regularisation (rate = 0.2) was applied after each LSTM layer to prevent overfitting. Random seeds were fixed across PyTorch, NumPy, and Python's random module to ensure fully reproducible results.

3 Analysis and Insights

3.1 Model Comparison

Table 7 and Figure 4 summarise the out-of-sample performance of all four models on the test set (July 2017 – December 2024).

Table 7: Out-of-Sample Forecast Performance (Test Set: July 2017 – December 2024)

Model	**MAE**	**RMSE**
ARIMAX(3,1,0)	1.752	2.836
LSTM	0.846	1.331
XGBoost (Baseline)	**0.641**	**1.146**
XGBoost (Tuned)	**0.641**	**1.146**
Improvement (XGBoost vs ARIMAX)	63.4%	59.6%
Improvement (XGBoost vs LSTM)	24.2%	13.9%

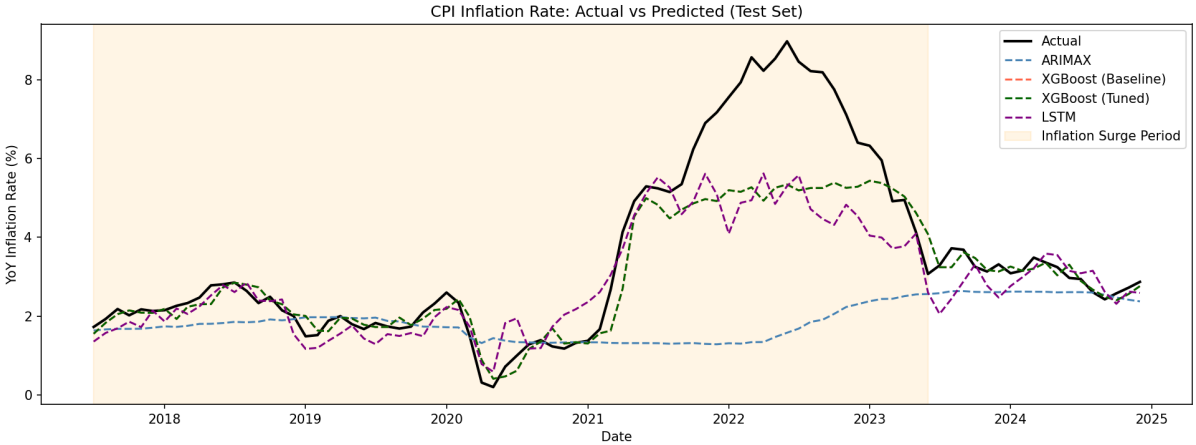

Figure 4: Actual vs. Predicted CPI Inflation Rate (Test Set: July 2017 – December 2024). The shaded orange region denotes the 2021–2023 inflation surge. XGBoost (red and green) tracks the actual series most closely, while ARIMAX (blue) consistently underestimates during the surge and LSTM (purple) captures directional trends but with greater variance.

XGBoost achieves the lowest MAE of 0.641 and RMSE of 1.146, outperforming ARIMAX by 63.4% and LSTM by 24.2% on MAE. A notable finding is that XGBoost Baseline and Tuned perform identically, as the GridSearchCV selected the same hyperparameters as the baseline (n_estimators = 200, max_depth = 3, learning_rate = 0.05), confirming that the initial parameter choices were already near-optimal.

3.2 Why Does XGBoost Outperform LSTM?

The superior performance of XGBoost over LSTM is an important and somewhat counterintuitive finding. Several factors explain this result.

First, while the 450-observation dataset is substantially larger than typical macroeconomic forecasting datasets, it may still be insufficient for LSTM to fully exploit its se-

quential learning capabilities. Deep learning models generally require thousands of observations to outperform well-tuned gradient boosting methods, and 348 training sequences is a relatively small sample for a two-layer LSTM.

Second, inflation forecasting is fundamentally driven by a small number of strong signals — most importantly, lagged inflation — that are already well-captured by XGBoost's non-linear interaction modelling. The additional complexity of LSTM's sequential architecture introduces variance that is not offset by a corresponding reduction in bias at this sample size.

Third, LSTM's training relied on early stopping rather than extensive hyperparameter optimisation. A more comprehensive architecture search could potentially improve LSTM's performance, though this would require significantly more computational resources.

3.3 Economic Interpretation

The dominance of `inflation_lag1` in both LASSO and XGBoost reflects the well-documented persistence of inflation, consistent with the concept of inflation inertia in New Keynesian Phillips Curve models. Firms and households form adaptive expectations, meaning current inflation is heavily anchored to recent experience.

The significance of the federal funds rate in ARIMAX (coef = 0.719, $p = 0.023$) may appear counterintuitive — higher interest rates are conventionally expected to reduce inflation, yet the coefficient is positive. This reflects *reverse causality*: the Fed raises rates in response to observed inflation, so the two series are positively correlated in the data even though the causal mechanism runs in the opposite direction. This is a well-known identification challenge in reduced-form macroeconomic models.

The insignificance of oil price changes in ARIMAX (p = 0.440) is noteworthy, despite oil_change receiving the third-highest LASSO coefficient (0.132). This divergence suggests that oil's predictive power for inflation is better captured through non-linear interactions with other variables — a role that XGBoost is better positioned to exploit than a linear ARIMAX specification.

The LASSO result that only M2 growth was eliminated is also informative. With the full 38-year dataset covering multiple monetary policy regimes, most macroeconomic variables carry meaningful predictive signal, in contrast to shorter-window analyses where many variables may appear redundant.

3.4 Performance During the Inflation Surge

All models underestimated the 2021–2023 inflation surge to some degree. This systematic underestimation reflects a genuine structural break: the surge was driven by pandemic-

induced supply chain disruptions, fiscal stimulus-driven demand shocks, and energy price spikes following the Russia-Ukraine conflict — dynamics not fully captured by the macroeconomic variables in our dataset. This illustrates a fundamental limitation of data-driven forecasting models: they extrapolate from historical patterns and cannot anticipate structural breaks driven by unprecedented events.

3.5 Limitations and Potential Improvements

Several limitations should be acknowledged. First, the dataset does not include supply-side indicators such as the Global Supply Chain Pressure Index (GSCPI) or shipping costs, which were critical drivers of the 2021–2023 surge. Second, all models use a fixed training window without re-estimation as new data arrive; a rolling or expanding window approach would better simulate a real-world forecasting environment. Third, the LSTM architecture was not extensively optimised; a more comprehensive hyperparameter search or transformer-based architecture could potentially improve its performance. Fourth, standard k-fold cross-validation violates the temporal ordering assumption, which may explain why XGBoost tuning produced identical results to the baseline.

Future improvements could include incorporating Federal Reserve communication sentiment using NLP techniques, applying regime-switching models to handle structural breaks explicitly, or combining ARIMAX and XGBoost predictions in an ensemble to leverage complementary strengths of statistical and machine learning approaches.

4 Coding and Reproducibility

All analysis was conducted in Python using the following libraries: `fredapi` for data retrieval from FRED, `sklearn` for LASSO feature selection, preprocessing, and GridSearchCV, `statsmodels` for ARIMAX estimation, `xgboost` for gradient boosting, `torch` (PyTorch) for LSTM implementation, and `matplotlib` and `seaborn` for visualisation. Random seeds were fixed at 42 across PyTorch, NumPy, and Python's random module to ensure fully reproducible LSTM results. The full notebook is structured sequentially and can be reproduced from top to bottom by running all cells in order. All data are downloaded programmatically via the FRED API, requiring no manual file downloads.

5 Conclusion

This project demonstrates that tree-based machine learning methods can substantially outperform both traditional time series models and deep learning approaches in forecasting U.S. CPI inflation on a moderately-sized macroeconomic dataset. XGBoost achieved an MAE of 0.641 on the July 2017 – December 2024 test set, compared to 0.846 for

LSTM and 1.752 for ARIMAX — a 63.4% improvement over the traditional baseline. The identical performance of XGBoost Baseline and Tuned confirms that the initial hyperparameter choices were already near-optimal, and that the inflation forecasting signal is well-structured for gradient boosting methods. While LSTM captures directional inflation dynamics reasonably well, its performance is constrained by the sample size available for sequential training. LASSO feature selection confirmed that inflationary inertia — as measured by lagged CPI inflation — is the single most informative predictor, consistent with adaptive expectations theory, while the retention of most macroeconomic variables highlights the broad predictive relevance of the full macroeconomic environment over long historical periods. All models struggled to fully anticipate the 2021–2023 inflation surge, reflecting the inherent difficulty of forecasting during structural break episodes driven by unprecedented supply-side shocks.